

ADNAN MASHRUR SADAD

AI / ML ENGINEER | APPLIED AI & ML SYSTEMS

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GitHub | LinkedIn | Portfolio

Software engineering graduate building evidence-grounded LLM applications, hybrid retrieval systems and deployable ML pipelines, with enterprise experience at Joget and first-author research in RAG and Text-to-SQL evaluation.

Technical Skills

AI & Data: Python, SQL, PyTorch, scikit-learn, XGBoost, pandas, NumPy, Hugging Face

Applied AI: RAG, hybrid retrieval, reranking, LLM evaluation, Text-to-SQL, structured-output validation

Applications & ML Systems: FastAPI, Pydantic, React, TypeScript, Groq, PostgreSQL/pgvector, Redis, Docker, Google Cloud, Kafka/Redpanda, PyFlink, MLflow, ONNX, GitHub Actions

Experience

Software Developer Intern | *Joget Inc.*

Oct 2024 - Mar 2025

- Built and deployed **Joget DX workflow automations and UI components** for client teams, working with senior engineers, QA, and product managers in Agile sprints.
- Integrated **REST APIs** to remove manual data re-entry and maintained application modules through recurring production release cycles.

Selected Projects

ProofHire | Evidence-Grounded AI Career Agent | GitHub

2026

- Built an evidence-grounded resume and cover-letter pipeline linking claims to source files and commit SHAs; combined **lexical, vector and skill retrieval with LLM reranking** to match job requirements.
- Implemented structured generation, a separate **claim-verification stage** and deterministic fact/metric checks, with bounded repair and removal of claims flagged as unsupported.
- Optimised indexed candidate retrieval and score fusion, reducing vector-plus-lexical query p50 from **399.34 ms** to **5.01 ms** on a local, single-user benchmark of **20,000 synthetic rows**.

Driftline | Streaming Fraud Detection and MLOps | GitHub

2026

- Replayed **590K transactions** through Redpanda/PyFlink; built GraphSAGE over a **605K-node graph** with unit-tested leakage controls. XGBoost PR-AUC: **0.4751 chronological** versus 0.6974 random split.
- Implemented drift-triggered retraining with a promotion gate: candidate PR-AUC **0.4802 versus 0.2811** incumbent on the same holdout. Served XGBoost via **ONNX/FastAPI on Google Cloud**: 10,018 requests, zero failures, **380 ms p95** at **50 concurrent users** on a 4-vCPU VM.

InterviewPilot | Applied AI Interview Platform | GitHub

2026

- Built a **React/FastAPI** app with persistent interviews, Groq transcription, schema-validated evaluation, bounded follow-ups, repair loops and disclosed fallback scoring. Verified **19 backend tests**, **89% statement coverage** and session-to-report CI browser checks; external AI calls mocked or disabled in tests.

Research

SAFE-RAG | Regulatory RAG Evaluation | GitHub

2025 - 2026

- First author; **short paper under review at ALTA 2026**. Evaluated answers with valid output schemas and supporting citations; identified an **8.6% wrong-scope grounding rate** across 200 annotated items.

Text-to-SQL Answerability under Schema Evolution

2026

- First author; manuscript complete, not yet submitted. Built a **1,382-instance benchmark across 20 databases** to evaluate answerability as database schemas change; ran 27,384 model generations.
- Detected **79.5% of structurally invisible drift at a 13.0% false-alarm rate**; evaluated against 5,627 benign updates.

Education

Universiti Teknologi Malaysia (UTM), MJIIT | Kuala Lumpur | 2021 - 2026

B.Sc. Computer Science (Software Engineering) with Honours | **CGPA: 3.43/4.00**

Dean's List: 2022/2023, 2025/2026.

Awards & Leadership

Gold Medal & Best Video Award, myHCI-UX Student Design Challenge (Oct 2025).

Secretary Treasurer, SOFEA Society (2022 - 2023); organised AI workshops for 200+ students.